Day 1: Properties of Exponents (Variable Exponents)

1. Simplify: $x^{n+3} \cdot x^{2n-1} =$ ____________
2. Simplify: $\dfrac{(3x^2y^{-3})^2}{9x^{-1}y^2} =$ ______________
3. **SAT:** If $2^{3x} = 2^{12}$, what is the value of x?

   A. 4 B. 9 C. 15 D. 36

4. If $a^{2x} \cdot a^3 = a^{15}$, what is the value of x? __________

Day 2: Rational Exponents & Radicals

1. **SAT:** Which is equivalent to $\sqrt{x} \cdot \sqrt[3]{x}$?

   A. $x^{1/6}$ B. $x^{2/5}$ C. $x^{5/6}$ D. $x^{6/5}$

2. Simplify: $(x^{1/3})^{-6} =$ ____________
3. What is the value of $16^{-3/4}$?

   A. -12 B. -8 C. $\frac{1}{8}$ D. 8

4. Simplify completely: $\sqrt[3]{54x^8y^{12}} =$ __________________

Day 3: Polynomial Operations

1. Simplify: $(5x^3 - 2x^2 + 4x - 1) - (2x^3 + 3x^2 - x + 7) =$ ________________________
2. Expand completely: $(x-2)(x+2)(x+3)$

   Show work: Answer: ________________________

3. **SAT:** Which is equivalent to $(x+1)^3$?

   A. $x^3 + 1$ B. $x^3 + 3x^2 + 3x + 1$ C. $x^3 + 3x + 1$ D. $x^3 + x^2 + x + 1$

Day 4: Multiply Polynomials + Box Method Preview

1. Expand: $(3x-2)(2x^2 - x + 5) =$ ______________________________
2. Use the box to multiply $(x+3)(x^2 - 2x + 4)$:

	x^2	$-2x$	$+4$
x			
$+3$			

Answer: ________________________

3. **SAT:** The expression $-x^2(-x)^4$ is equivalent to:

   A. x^6 B. $-x^6$ C. x^8 D. $-x^8$

Day 5: Division + Mixed Review

1. Simplify: $\dfrac{18x^5y^3 - 12x^3y^4 + 6x^2y^2}{6x^2y^2} =$ ___________________________

2. **SAT:** The expression $\dfrac{(3y^3)(2y^2)^2}{(y^4)^3}$ is equivalent to:

   A. $\dfrac{12}{y^5}$ B. $12y^5$ C. $\dfrac{6}{y^5}$ D. $12y^{-5}$

3. **SAT:** If $\sqrt[3]{b^y} = b^5$, what is the value of y?

   A. $\dfrac{5}{3}$ B. 5 C. 8 D. 15

4. Simplify: $(27x^6)^{2/3} =$ ________________